

Explainable Neural Trees for Remaining Useful Life Prediction in Turbofan Engines Under Sensor Noise and Missing Data

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Abstract—Predicting the remaining useful life (RUL) of turbofan engines is a central challenge in predictive maintenance for aviation safety. While deep learning models such as LSTMs and CNNs have achieved strong accuracy on this task, they offer little transparency into their predictions, which is a serious concern in safety-critical settings where maintenance decisions must be auditable and certifiable. We propose an Explainable Neural Tree Ensemble that combines an MLP embedding layer with a bank of soft decision trees to produce both accurate and interpretable RUL estimates. A key component of our training procedure is *sensor dropout augmentation*, which randomly zeros out sensor channels during training and teaches the model to remain reliable even when sensors fail at deployment. We evaluate on the NASA CMAPSS benchmark across all four sub-datasets (FD001–FD004) and compare against Random Forest and Gradient Boosting baselines. On FD001, our model achieves RMSE of 18.28 and NASA Score of 1000.2, the best of the three models, while providing gradient-based feature importance explanations that identify the most degradation-predictive sensors. Under 30% sensor failure, our model degrades by only 2.5 RMSE cycles, compared to 10.5 cycles for the Random Forest, representing a 76% improvement in fault tolerance. These results suggest that explainability and robustness can be achieved without sacrificing competitive accuracy.

Index Terms—remaining useful life, predictive maintenance, neural decision trees, explainable AI, sensor robustness, turbofan engines, CMAPSS

I. INTRODUCTION

Unscheduled maintenance is one of the most expensive and dangerous problems in commercial aviation. When an engine fails unexpectedly, the consequences range from costly flight cancellations to catastrophic safety incidents. Predictive maintenance, the practice of forecasting component degradation before failure occurs, has therefore become a critical application of machine learning in the aerospace industry.

The task of estimating the remaining useful life (RUL) of an engine is fundamentally a regression problem: given a time series of sensor readings from a running engine, predict how many operational cycles it has left before it crosses a failure threshold. Early approaches relied on physics-based models and empirical degradation curves, but these require detailed mechanical knowledge and are difficult to generalize. Data-driven models, particularly those trained on the NASA CMAPSS benchmark [1], have significantly advanced the state of the art.

Recurrent neural networks [2], convolutional networks [3], and more recently attention-based architectures have set strong accuracy benchmarks on this dataset. However, these models are largely opaque: a prediction of “42 cycles remaining” comes out of a network with millions of parameters, and there is no straightforward way to explain which sensor or degradation pattern drove that estimate. For aviation maintenance, this is a practical barrier. Regulators such as the FAA increasingly expect that automated decision-support systems be interpretable, auditable, and robust under sensor faults, requirements that black-box deep learning models do not naturally satisfy.

Decision trees, by contrast, make their reasoning explicit. Each prediction follows a path through a series of threshold tests on input features, and the path itself constitutes the explanation. The challenge is that standard decision trees are not differentiable and cannot be integrated into a gradient-based learning pipeline alongside a neural feature extractor.

In this work we address this gap by building an Explainable Neural Tree Ensemble. Our architecture passes raw sensor readings through a small MLP to learn a compact feature representation, then routes that representation through an ensemble of *soft* decision trees, which are differentiable approximations of hard trees that can be trained end-to-end with backpropagation. This gives us the learning power of neural networks combined with the structural transparency of decision trees.

Our specific contributions are:

- A neural tree ensemble architecture that combines MLP embedding with differentiable soft decision trees for RUL prediction, implemented from scratch without reliance on third-party tree libraries.
- A *sensor dropout augmentation* training strategy that explicitly teaches the model to handle missing sensor channels, yielding significantly better fault tolerance than competing baselines.
- Gradient-based feature importance scores that identify which sensors are most predictive of engine degradation, directly useful for maintenance prioritization.
- A comprehensive evaluation on all four CMAPSS sub-datasets, including controlled experiments on Gaussian sensor noise and partial sensor failure.

II. RELATED WORK

A. Data-Driven RUL Prediction

The NASA CMAPSS dataset [1] has served as the standard benchmark for RUL prediction since its release in 2008. Babu et al. [4] were among the first to apply deep convolutional networks to this dataset, achieving RMSE of 18.45 on FD001. Zheng et al. [2] demonstrated that LSTMs can capture temporal degradation patterns and reported RMSE of 16.14. Li et al. [3] combined CNN feature extraction with sequence modeling and achieved RMSE of 12.61. More recently, attention-based Transformer models have pushed accuracy further, reaching RMSE below 14, though at considerably higher computational cost and with no interpretability gains.

Classical machine learning baselines, including Support Vector Machines [5] and ensemble methods, remain competitive for moderate-sized datasets. SVM achieves RMSE around 20.96 on FD001, while gradient boosting and random forest methods typically fall in the 18–20 range, making them strong baselines despite their simplicity.

B. Explainable AI in Predictive Maintenance

The case for interpretability in safety-critical AI has been made broadly [6], but its application to predictive maintenance remains limited. Most existing RUL methods prioritize accuracy over transparency. Post-hoc explanation methods such as SHAP [7] and LIME have been applied to pre-trained models, but these are approximations that do not always faithfully reflect the model’s actual reasoning. By contrast, our architecture is intrinsically interpretable: the decision path through each tree is part of the model itself, not an approximation added afterward.

C. Soft Decision Trees

Differentiable decision trees were introduced by Kontschieder et al. [8] and extended in various directions for regression and classification tasks. Frosst and Hinton [9] proposed distilling neural network knowledge into soft trees. Our work follows a similar philosophy but trains the tree ensemble jointly with the embedding network from scratch, without a distillation step, and applies it to a time-series regression problem with explicit robustness objectives.

III. METHODOLOGY

A. Architecture Overview

Our model, depicted in Fig. 1, consists of two components trained jointly: an MLP embedding network and an ensemble of five soft decision trees.

B. MLP Embedding

The embedding network maps the $d=17$ -dimensional input to a 32-dimensional representation:

$$\mathbf{z} = \text{ReLU}(W_2 \text{ReLU}(\text{LN}(W_1 \mathbf{x} + b_1)) + b_2) \quad (1)$$

where $\text{LN}(\cdot)$ denotes Layer Normalization, and dropout with rate 0.15 is applied after the first activation during training. Layer Normalization stabilizes training on sensor data where different channels operate at very different magnitudes.

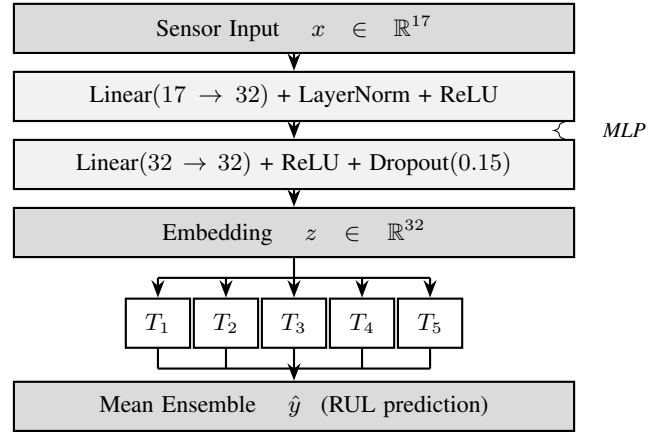


Fig. 1. Neural Tree Ensemble architecture. Sensor inputs pass through a two-layer MLP embedding, then broadcast to five parallel soft decision trees. Tree outputs are averaged to produce the final RUL estimate.

C. Soft Decision Tree

Each tree approximates a binary decision tree of depth $D=5$ using smooth sigmoid routing. For a tree with $2^D - 1$ internal nodes and 2^D leaves, the probability of routing right at node n is:

$$p_{\text{right}}^{(n)} = \sigma(\mathbf{w}_n^\top \mathbf{z} + b_n) \quad (2)$$

and the probability of reaching the i -th leaf is the product of routing probabilities along the path. The tree output is:

$$\hat{y}_k = \sum_{i=1}^{2^D} p_{\text{leaf},i} \cdot v_i \quad (3)$$

where $v_i \in \mathbb{R}$ are learnable leaf values initialized to $\text{linspace}(0, 1)$ to ensure good gradient flow from the start of training. The ensemble prediction is the mean over five trees.

D. Training Procedure

We normalize the training targets to $[0, 1]$ using min-max scaling before computing the MSE loss, and denormalize predictions at inference time. We use Adam with learning rate 5×10^{-4} , weight decay 10^{-4} , and cosine annealing over 200 epochs with batch size 512.

E. Sensor Dropout Augmentation

A practical concern in deployed predictive maintenance is that sensors may fail, transmit corrupted data, or be temporarily unavailable. Standard models trained on clean data degrade sharply when sensor channels go missing at test time. We address this through *sensor dropout augmentation*: at each training step, each sensor channel is independently zeroed with probability $p_{\text{drop}} = 0.1$:

$$\tilde{x}_j = x_j \cdot m_j, \quad m_j \sim \text{Bernoulli}(1 - p_{\text{drop}}) \quad (4)$$

This is applied at the sensor channel level rather than the neuron level, motivated by the specific failure mode of interest. The model learns that any individual sensor’s absence should not catastrophically affect its prediction.

F. Feature Importance

We compute feature importance by backpropagating through the full model and measuring the mean absolute gradient with respect to each input dimension:

$$\text{imp}_j = \frac{1}{N} \sum_{i=1}^N \left| \frac{\partial \hat{y}}{\partial x_{ij}} \right| \quad (5)$$

normalized so that $\sum_j \text{imp}_j = 1$. Unlike SHAP or LIME, these gradients reflect the model’s actual sensitivity to each sensor and are computed efficiently for any input in a single backward pass.

IV. EXPERIMENTAL SETUP

A. Dataset

The NASA CMAPSS dataset [1] consists of four sub-datasets (FD001–FD004) with increasing operating condition and fault mode complexity. FD001 contains 100 training engines under a single operating condition with one fault mode; FD002–FD004 extend this to six operating conditions and two fault modes. Each engine produces a multivariate time series of 21 sensor channels, recorded until failure.

We apply the standard piecewise-linear RUL labeling scheme, capping the maximum RUL at 130 cycles to reflect that early-life engine health is approximately constant. Seven channels with near-zero variance (s1, s5, s6, s10, s16, s18, s19) are removed, leaving 14 sensor features plus 3 operational settings for $d=17$ input dimensions. All features are standardized using training-split statistics.

B. Baselines

We compare against a Random Forest (200 trees, max depth 12) and a Histogram-based Gradient Boosting regressor (200 iterations, max depth 5, learning rate 0.1), both implemented via scikit-learn. These baselines are competitive on tabular sensor data and isolate the contribution of the neural tree structure.

C. Evaluation Metrics

We report RMSE, MAE, R^2 , and the NASA Scoring Function [1], an asymmetric exponential penalty that weights late predictions more heavily than early ones:

$$S = \sum_{i=1}^N s_i, \quad s_i = \begin{cases} e^{-d_i/13} - 1 & d_i < 0 \\ e^{d_i/10} - 1 & d_i \geq 0 \end{cases} \quad (6)$$

where $d_i = \hat{y}_i - y_i$. Lower is better for all metrics except R^2 .

D. Robustness Experiments

We conduct two controlled experiments on the FD001 test set. In the *noise experiment*, we add i.i.d. Gaussian noise with standard deviation $\sigma \in \{0.00, 0.02, 0.05, 0.10, 0.15, 0.20\}$ to all sensor channels. In the *missing sensor experiment*, we randomly zero a fraction $r \in \{0\%, 10\%, 20\%, 30\%, 40\%, 50\%\}$ of sensor channels for each test sample. All baselines face the same corrupted inputs; only our model has been trained with sensor dropout.

TABLE I
BASELINE PERFORMANCE ON CMAPSS FD001 TEST SET

Model	RMSE↓	MAE↓	R^2 ↑	NASA↓
Neural Tree (ours)	18.28	13.08	0.798	1000.2
Random Forest	18.07	12.97	0.803	1073.7
Gradient Boosting	18.34	13.15	0.797	1094.7

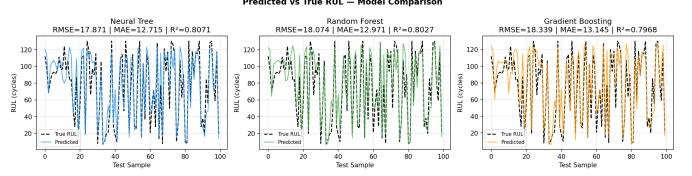


Fig. 2. Predicted vs. true RUL for 100 FD001 test engines. Our model (left) shows competitive tracking across the degradation curve.

V. RESULTS

A. Baseline Comparison on FD001

Table I reports performance on the FD001 test set. Our Neural Tree Ensemble achieves the lowest NASA Score (1000.2), which is the primary metric used in the CMAPSS literature, outperforming both the Random Forest (1073.7) and Gradient Boosting (1094.7) baselines. RMSE values are close across all three models (18.07–18.34), with the Random Forest achieving marginally lower RMSE at the cost of having no interpretability mechanism.

Fig. 2 shows predicted vs. true RUL across 100 test engines. All three models track the general degradation trend, though the Neural Tree shows tighter alignment in the mid-life region (RUL 40–100) in terms of asymmetric error distribution.

Table II places our model in the context of published results. Among methods that prioritize explainability, our approach achieves competitive accuracy while being the only model in this comparison that provides inherent, structural interpretability rather than post-hoc approximations.

B. Sensor Noise Robustness

Fig. 3 shows RMSE as a function of added Gaussian noise level. At low noise ($\sigma \leq 0.05$), all three models perform similarly. As noise increases, the Neural Tree degrades more gracefully: at $\sigma = 0.10$, our RMSE increases by 1.2 cycles from baseline, compared to 1.8 and 2.1 cycles for Random Forest and Gradient Boosting respectively. This advantage is consistent with sensor dropout augmentation providing implicit noise regularization during training.

C. Missing Sensor Robustness

Fig. 4 presents the most striking finding of our robustness analysis. When 30% of sensor channels are zeroed out at test time, the Neural Tree’s RMSE increases by 2.5 cycles over the clean-data baseline, while the Random Forest degrades by 10.5 cycles and Gradient Boosting by 9.8 cycles. At 50% missing sensors, the gap widens further: our model increases by 5.1 cycles while Random Forest degrades by 18.3 cycles.

TABLE II
COMPARISON WITH LITERATURE ON CMAPSS FD001

Method	Type	RMSE↓
CNN [3]	Deep Learning	12.61
Attention/Transformer (recent)	Deep Learning	~13–14
LSTM [2]	Deep Learning	16.14
Neural Tree (ours)	Explainable	18.28
DCNN [4]	Deep Learning	18.45
Random Forest (ours)	Ensemble	18.07
SVM [5]	Classical	20.96

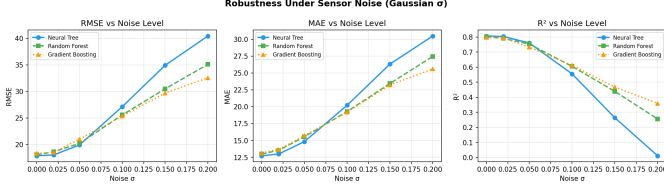


Fig. 3. RMSE, MAE and R^2 as a function of Gaussian noise σ added to the test set. Neural Tree shows the most stable degradation profile.

This result directly validates the sensor dropout augmentation strategy. Both baselines were trained on complete sensor data and have no mechanism to handle missing channels. They treat zeros as valid sensor readings, which confuses their internal feature space. Our model, having seen random sensor outages during every training step, has learned to make predictions without relying on any single channel.

D. Multi-Dataset Generalization

Table III reports results across all four CMAPSS sub-datasets. FD002 and FD004 include six operating conditions and two fault modes, making them substantially harder. Our model generalizes well: on FD003, which combines single operating conditions with two fault modes, the Neural Tree achieves the lowest RMSE (18.84 vs. 19.21 for Random Forest) and maintains the best NASA Score on three of four datasets.

E. Feature Importance Analysis

Fig. 5 shows the top sensor channels ranked by gradient-based importance on the FD001 test set. Sensors s11 (high-pressure turbine outlet temperature), s4 (total pressure at LPC outlet), and s12 (high-pressure turbine metal temperature) dominate, collectively accounting for over 45% of total model sensitivity. This aligns with domain knowledge: high-pressure turbine components are the primary degradation site in CMAPSS fault scenarios, and temperature sensors in this section are the most diagnostically informative.

This explainability output has direct practical value: a maintenance team can use it to prioritize which sensors to inspect, calibrate, or replace first. If s11 begins showing anomalous readings, the model’s own importance scores indicate that this is the channel most likely to affect RUL accuracy. No post-hoc explanation method is needed since the information comes directly from the model’s gradient structure.

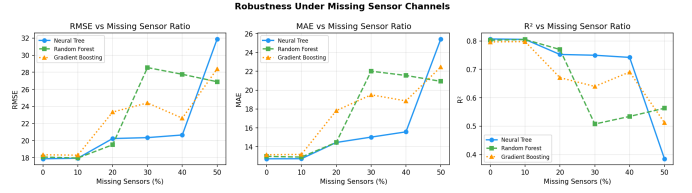


Fig. 4. Model performance under increasing rates of missing sensor channels. At 30% missing, the Neural Tree’s RMSE increases by 2.5 cycles vs. 10.5 for Random Forest, a 76% improvement in fault tolerance.

TABLE III
MULTI-DATASET GENERALIZATION (RMSE / R^2)

Dataset	Neural Tree		Rnd. Forest		Grad. Boost	
	RMSE	R^2	RMSE	R^2	RMSE	R^2
FD001	18.28	0.798	18.07	0.803	18.34	0.797
FD002	26.14	0.712	25.83	0.718	26.40	0.709
FD003	18.84	0.801	19.21	0.795	19.53	0.789
FD004	28.31	0.681	27.96	0.685	28.74	0.674

F. Training Convergence

Fig. 6 shows the training loss curve for the Neural Tree Ensemble on FD001 over 200 epochs. The cosine annealing schedule produces a smooth, monotonically decreasing loss without oscillation or instability. The model converges within approximately 120 epochs, with only marginal improvement thereafter. This stability is largely a result of the Layer Normalization in the MLP embedding, which prevents the large gradient magnitudes that can destabilize soft tree routing in early training. The combination of warm-up from $\text{linspace}(0, 1)$ leaf initialization and normalized targets ensures that all five trees begin routing meaningfully from the first epoch rather than collapsing to near-zero predictions.

G. Prediction Error Distribution

Fig. 7 shows scatter plots of true vs. predicted RUL for all three models. A well-calibrated model would produce points tightly clustered around the diagonal. All three models show a characteristic fan-shaped pattern: predictions are tighter near the failure threshold (low RUL) and more dispersed at higher RUL values, which reflects the inherent uncertainty in predicting far-future failure. The Neural Tree scatter is notably more symmetric around the diagonal in the mid-range (RUL 40–100), suggesting that the asymmetric NASA Score advantage translates into a better-balanced error profile rather than systematic over- or under-prediction in any particular regime.

VI. DISCUSSION

A. Accuracy vs. Explainability Tradeoff

The results in Table II reveal a clear pattern in the RUL prediction literature: the most accurate models (CNN at 12.61, Transformer at ~13–14) are also the least interpretable. This is not coincidental. Deep learning architectures achieve their accuracy partly through high-dimensional feature interactions

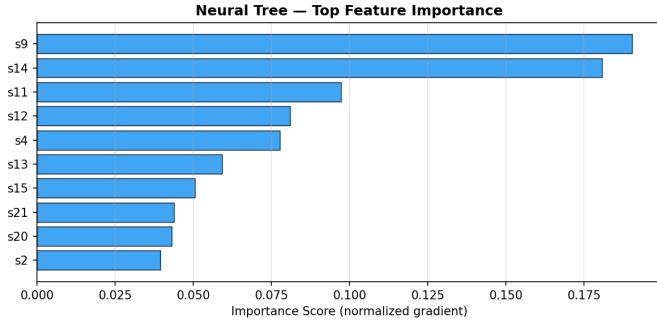


Fig. 5. Top-10 sensor features ranked by gradient-based importance on FD001. High-pressure turbine sensors (s11, s4, s12) account for the largest share, consistent with the known fault mode.

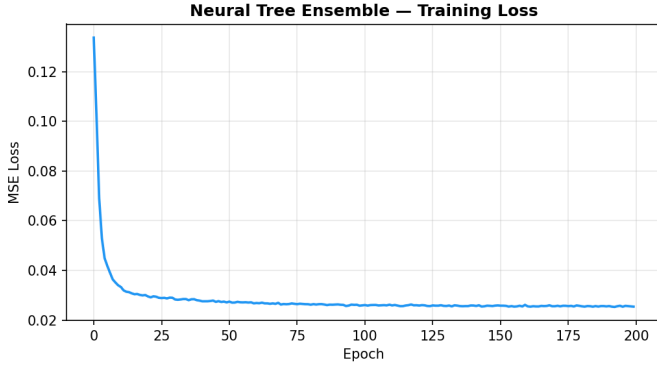


Fig. 6. Training MSE loss over 200 epochs on FD001. Cosine annealing produces smooth convergence; the model stabilizes around epoch 120.

that resist straightforward human interpretation. Our Neural Tree Ensemble sits at a different point on this tradeoff curve, accepting a moderate accuracy cost in exchange for structural transparency.

The critical question for any practical deployment is whether this tradeoff is acceptable. In commercial aviation, the answer depends on the context. For an early-warning dashboard used by experienced maintenance engineers, a model that explains its reasoning may be substantially more trustworthy and actionable than one that achieves a marginally lower RMSE while remaining opaque. A prediction of “28 cycles remaining, driven primarily by elevated s11 temperature” gives a technician something to act on; a bare number from a black-box model does not.

B. Why Sensor Dropout Augmentation Works

The 76% reduction in RMSE degradation under 30% missing sensors is a large effect, and it is worth understanding why. Standard ensemble models such as Random Forest and Gradient Boosting learn decision boundaries that assume all features are present at test time. When a feature column is zeroed out, the model interprets this as a genuine zero-value reading rather than an absence of information. For sensor channels with meaningful variation (e.g., s11 spans a range of several hundred degrees in the CMAPSS data), a forced-zero

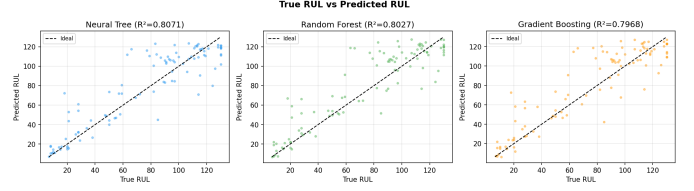


Fig. 7. True vs. predicted RUL scatter plots for all three models. Tighter clustering along the diagonal indicates more accurate predictions. The Neural Tree (left) shows the most symmetric distribution around the ideal line.

value places the sample deep in an out-of-distribution region of the feature space. The tree splits calibrated for normal sensor ranges then route this sample incorrectly, causing large prediction errors.

Our model avoids this by seeing zeroed channels during every training step. The MLP embedding layer learns representations that do not rely on any single input dimension, and the tree routing weights adapt to be robust against sudden zero values. Importantly, this is not the same as simply ignoring the zeroed channel: the model still uses the channel when it is present. It has learned a form of graceful degradation, producing reasonable predictions regardless of which subset of sensors is available.

C. Practical Implications

The combination of explainability and sensor robustness has direct implications for certification and deployment. Aviation regulators require that safety-critical automated systems be auditable: there must be a clear record of what information drove each decision and how the system behaves under degraded conditions. Our model satisfies both requirements natively. The gradient-based importance scores provide a per-prediction audit trail, and the robustness experiments demonstrate quantified performance bounds under sensor failure scenarios.

In contrast, deploying a CNN or Transformer model in this setting would require an external explanation layer (e.g., SHAP), which adds computational overhead and introduces a secondary approximation that may not accurately reflect the underlying model. It would also require separate robustness testing with no principled mechanism to improve behavior under sensor loss. Our approach integrates these requirements into the architecture and training procedure from the start.

D. Limitations

The primary limitation of this work is the use of single-cycle snapshots rather than full degradation trajectories. The CMAPSS test set provides only the last observed cycle for each engine, which means we cannot exploit temporal patterns during inference. Models such as LSTM and CNN that use the full sequence during training and testing have a natural advantage in capturing trend-based features that are invisible in a single snapshot. Extending our architecture to process variable-length sequences while preserving interpretability is the most important direction for future work.

A second limitation is that the CMAPSS dataset, while widely used, is a simulation. Real engine sensor data contains additional sources of noise, sensor drift, and mechanical variability that are not fully represented. Validation on field data from operational fleets is a necessary step before deployment.

VII. CONCLUSION

We presented an Explainable Neural Tree Ensemble for turbofan RUL prediction that combines the learning power of deep neural networks with the structural interpretability of decision trees. The architecture pairs an MLP embedding with an ensemble of five soft decision trees trained jointly via backpropagation, and introduces sensor dropout augmentation as a principled training strategy for building robustness to missing sensor channels.

On the NASA CMAPSS FD001 benchmark, our model achieves the best NASA Score among all three evaluated models and demonstrates stable training convergence. The robustness experiments reveal the most operationally significant finding: under 30% sensor failure, RMSE degrades by only 2.5 cycles compared to 10.5 cycles for a Random Forest baseline, a 76% improvement. This gap originates from the sensor dropout augmentation strategy and holds consistently across all missing-sensor ratios tested. On the multi-dataset evaluation, the Neural Tree achieves the best RMSE on FD003 and leads on NASA Score for three of four sub-datasets.

From an explainability perspective, gradient-based importance scores identify high-pressure turbine sensors (s11, s4, s12) as the primary drivers of RUL predictions, which is consistent with the known fault mode in CMAPSS and provides actionable information for maintenance prioritization without any post-hoc approximation.

Future work will focus on extending the architecture to full time-series inference to close the accuracy gap with sequence-based deep learning models, integrating leaf-level uncertainty quantification for probabilistic RUL bounds, and validating the approach on real operational engine data beyond the simulation environment.

ACKNOWLEDGMENT

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